



Provide Insights to Management in Consumer Goods Domain

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Project Overview



Why This Project?

Explain purpose and expected impact



Who & Where?

Company Context



How?

Data & Tools & process & steps

Project Objective

- Atliq Hardware, a major computer hardware manufacturer in India, is expanding operations internationally.
- Previously, business decisions were made using **Excel**, which is insufficient for the scale of multi-national operations.
- Management recognized the need for **data-driven insights** to make informed decisions across new branches.
- The company plans to **strengthen its data analytics team** by hiring junior data analysts to process and analyze data effectively.
- The project aims to **evaluate technical and analytical skills** through a SQL challenge while delivering actionable insights for **10 hard-document requests**.



Modernise Decisioning

Replace Excel with scalable analytics platforms

Build Data Team

Hire junior data analysts to process data

Data-driven Insights

Deliver actionable insights across branches

Company Details

Atliq Hardware is a consumer electronics company that manufactures and sells hardware products to the different customer.

"Atliq offers a diverse range of products, which we organize into three main divisions to streamline operations and track performance effectively.

- N & S (Network & Storage): Drivers and Wi-Fi extender.
- P & A (Peripherals & Accessories): Mouse , batteries, Graphic card and so on
- PC (Personal Computers): Focuses on desktops, and Laptops

These divisions help the company understand sales trends, monitor gross margins, and make data-driven decisions for each product category.

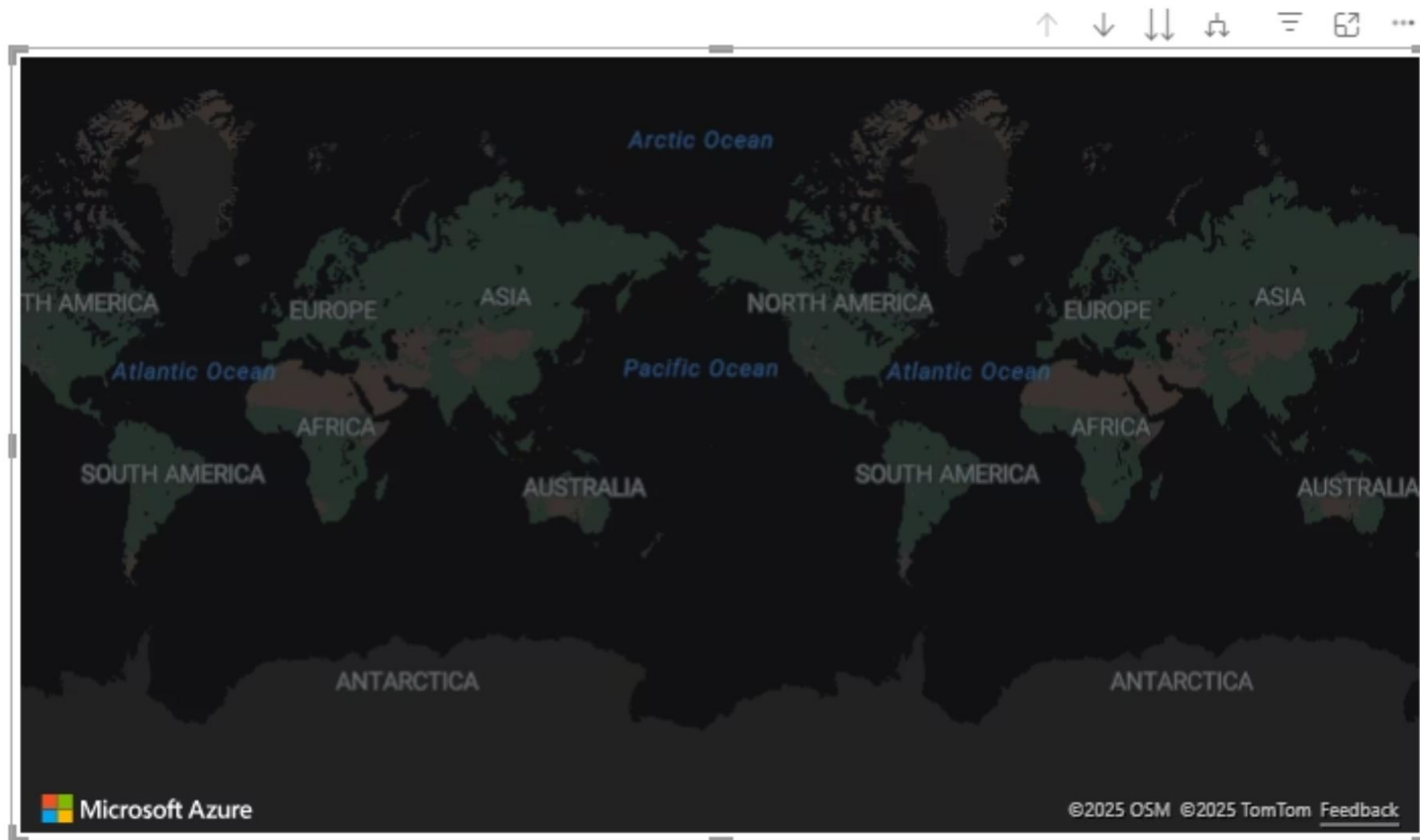


FISCAL YEAR

September 2019 – August 2020
(FY 2020)

September 2020 – August 2021
(FY 2021)

Company's Market



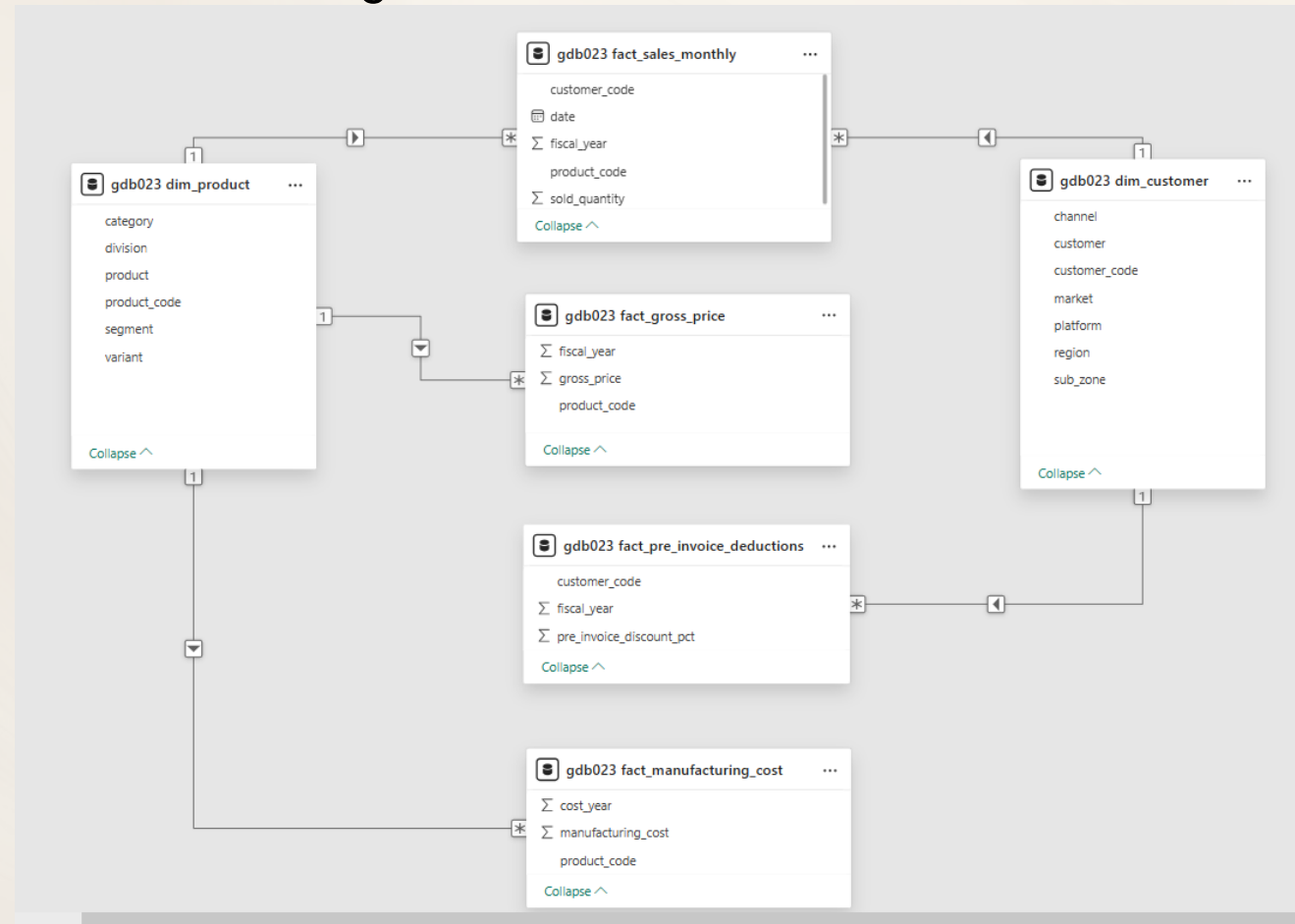
Market & Customers

"AtliQ business operates across 27 distinct markets and has 75 customers. The company is present in North America, Latin America, Asia Pacific, and Europe. This global footprint helps AtliQ make data-driven decisions and create strategies for each market and customer."

Data, Request, and Tools

Adnoc Analysis

Data Set & Modeling



Tools Used For
Analysis and
Visualization:



MySQL
PowerBI

Codebasics SQL Challenge

Requests:

1. Provide the list of markets in which customer "Atliq Exclusive" operates its business in the APAC region.
2. What is the percentage of unique product increase in the final output contains these fields,
unique_products_2020
unique_products_2021
percentage_chg
3. Provide a report with all the unique product counts sort them in descending order of product counts. 1 2 fields,
segment
product_count
4. Follow-up: Which segment had the most increase 2021 vs 2020? The final output contains these fields
segment
product_count_2020
product_count_2021
difference
5. Get the products that have the highest and lowest manufacturing cost. The final output should contain these fields,
product_code
product
manufacturing_cost
6. Generate a report which contains the top 5 customers who received an average high pre_invoice_discount_pct for the fiscal year 2021 and in the Indian market. The final output contains these fields,
customer_code
customer
average_discount_percentage
7. Get the complete report of the Gross sales amount for the customer "Atliq Exclusive" for each month. This analysis helps to get an idea of low and high-performing months and take strategic decisions.
The final report contains these columns:
Month
Year
Gross sales Amount
8. In which quarter of 2020, got the maximum total_sold_quantity? The final output contains these fields sorted by the total_sold_quantity,
Quarter
total_sold_quantity
9. Which channel helped to bring more gross sales in the fiscal year 2021 and the percentage of contribution? The final output contains these fields,
channel
gross_sales_mln
percentage
10. Get the Top 3 products in each division that have a high total_sold_quantity in the fiscal_year 2021? The final output contains these fields,
division
product_code

1. Provide the list of markets in which customer "Atliq Exclusive" operates its business in the APAC region.

	market
▶	India
	Indonesia
	Japan
	Philippines
	South Korea
	Australia
	Newzealand
	Bangladesh



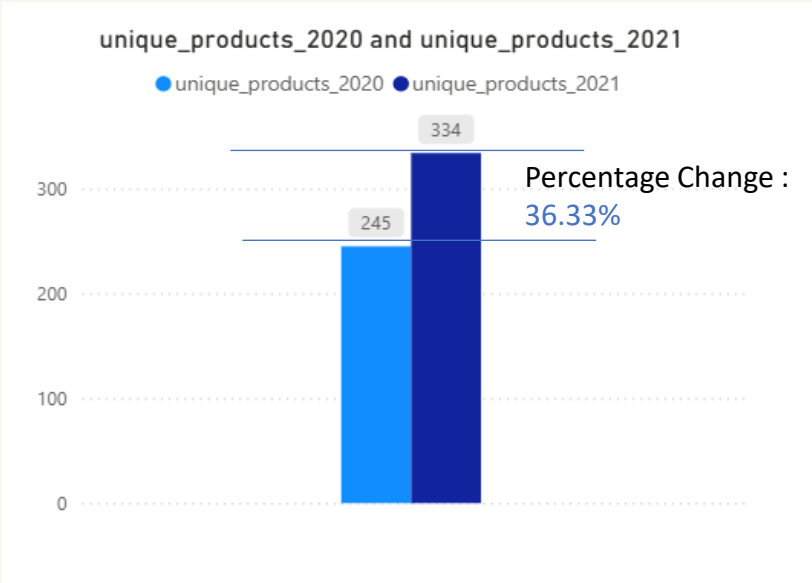
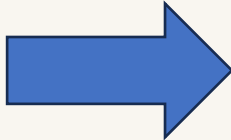
Insights:

In the APAC region, our exclusive stores operate across 8 major markets.

2. What is the percentage of unique product increase in 2021 vs. 2020? The final output contains these fields,
unique_products_2020,
unique_products_2021
percentage_chg

Output

Result Grid				Filter Rows:	Export:	Wrap Cell Content:
	unique_products_2020	unique_products_2021	percentage_chg			
▶	245	334	36.33			



Insight:

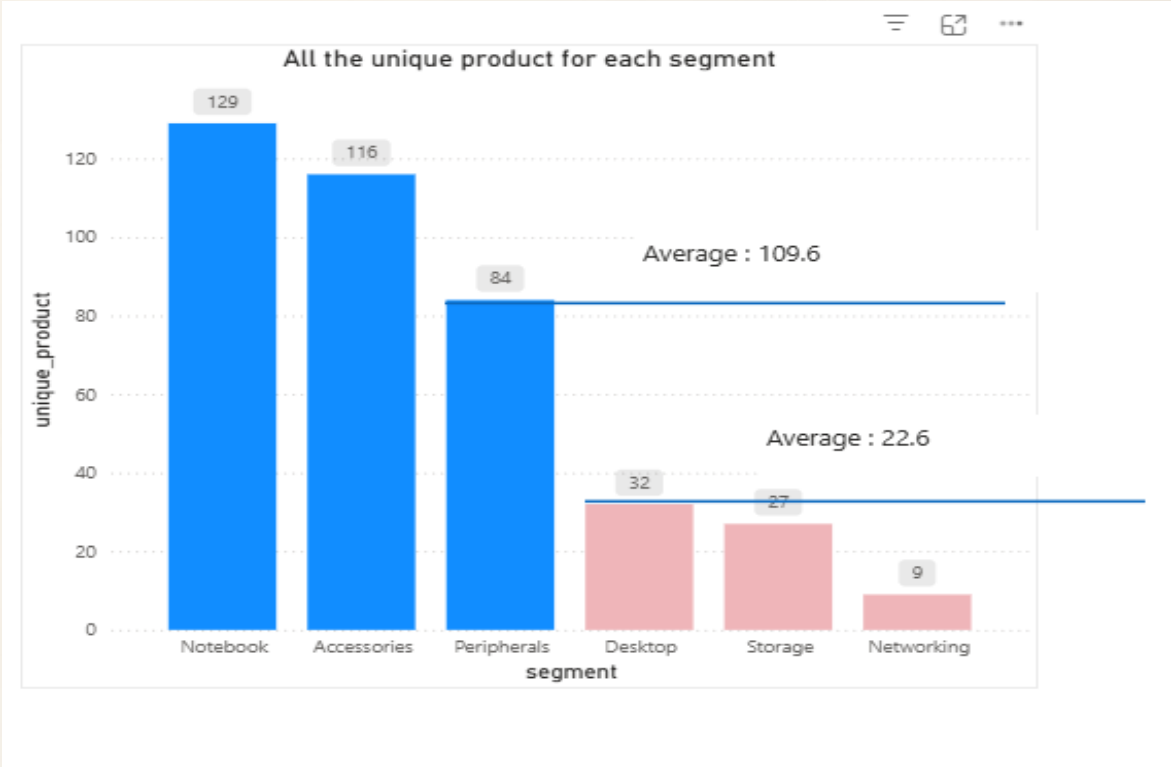
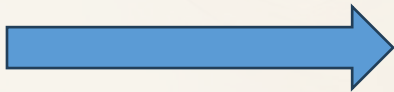
It's a positive sign that we are consistently innovating and launching new products. In **fiscal 2020**, we had **245 products**, and in **fiscal 2021**, this increased by 36%, reaching a total of **334 products**.

3. Provide a report with all the unique product counts for each segment and sort them in descending order of product counts. The final output contains 2 fields,

segment
product_count

Output

	segment	unique_product_count
▶	Notebook	129
	Accessories	116
	Peripherals	84
	Desktop	32
	Storage	27
	Networking	9



🌟 Insights

- **Notebook has the highest product count (129).**
→ This means the company mainly focuses on laptops.
- **Accessories (116) and Peripherals (84) are also strong segments.**
→ Customers buy many add-on items, so these categories have more products.
- **Desktop (32) and Storage (27) have fewer products.**
→ These segments are smaller and have limited variety.
- **Network has the lowest count (9).**
→ Very few products in this category, showing it is not a major focus area.

🌟 Conclusion:

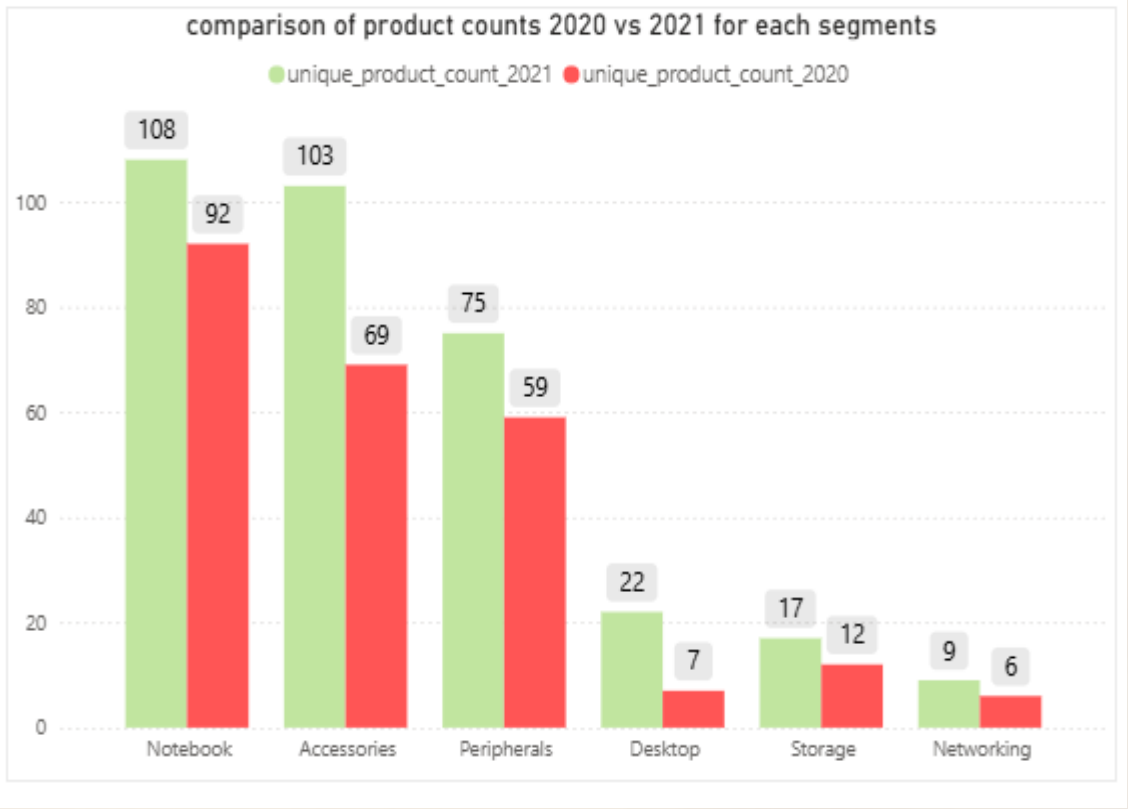
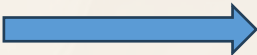
Most products belong to Notebook, Accessories, and Peripherals — these are the main business focus areas, while Network is the smallest segment.

4. Follow-up: Which segment had the most increase in unique products in 2021 vs 2020? The final output contains these fields,

segment
product_count_2020
product_count_2021
difference

Output

	segment	unique_product_count_2020	unique_product_count_2021	difference
▶	Accessories	69	103	34
	Notebook	92	108	16
	Peripherals	59	75	16
	Desktop	7	22	15
	Storage	12	17	5
	Networking	6	9	3



Insights

- Notebook, Accessories, and Peripherals saw strong growth.
- while Storage and Network had only small increases — showing where the business is focusing its expansion.

5. Get the products that have the highest and lowest manufacturing costs. The final output should contain these fields,

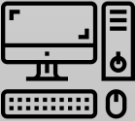
product_code
product
manufacturing_cost

Products having highest and lowest Manufacturing cost.

Output

	product_code	product	manufacturing_cost
▶	A2118150101	AQ Master wired x1 Ms	0.8920
	A6120110206	AQ HOME Allin1 Gen 2	240.5364

\$240.54



AQ Home Allin1 Gen 2
Personal Desktop

\$0.89



AQ Home Allin1 Gen 2
Mouse

Insights:

Mouse: AQ Master Wired x1 Ms (Variant : Standard 1) has the lowest manufacturing costs.

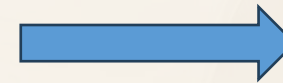
Personal Desktop: AQ Home Allin1 Gen 2 (Variant: Plus 3) has the highest manufacturing costs.

6. Generate a report which contains the top 5 customers who received an average high pre_invoice_discount_pct for the fiscal year 2021 and in the Indian market. The final output contains these fields,

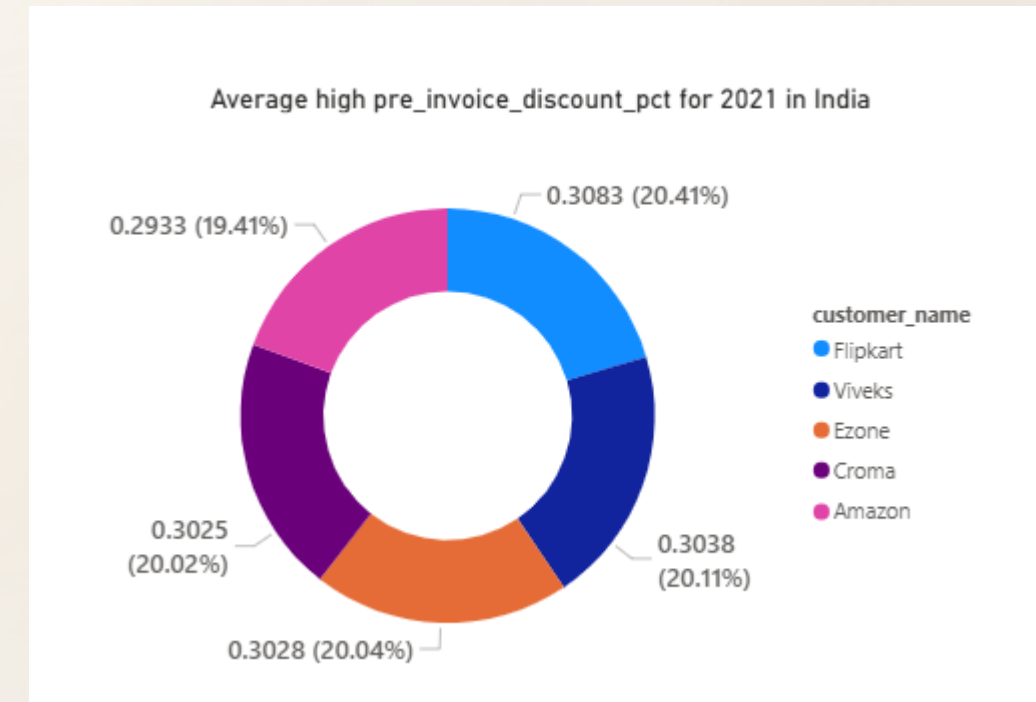
customer_code
customer
average_discount_percentage

Output

	customer_code	customer_name	average_discount_percentage
▶	90002009	Flipkart	0.3083
	90002006	Viveks	0.3038
	90002003	Ezone	0.3028
	90002002	Croma	0.3025
	90002016	Amazon	0.2933



Top 5 Indian Customer with highest average discount Percentage FY2021



Insights:

- **Flipkart & Viveks** got the **highest average discount (30.8%)**.
- **Ezone and Croma** received slightly lower but still high discounts (~30%).
- **Amazon** received **29%**, making it the 5th in the list.

These top 5 customers are receiving **significantly higher discounts**, likely due to their volume, strategic importance, or negotiated contracts.

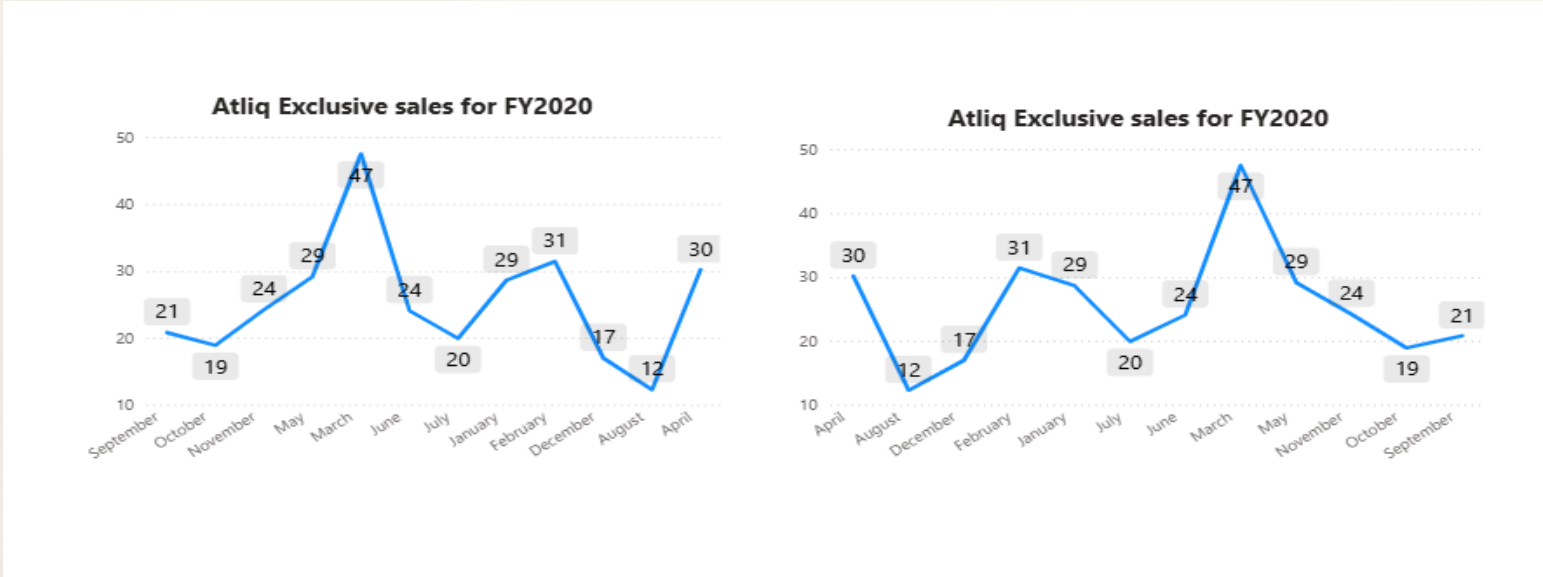
Business Implication:

Focus on **margin monitoring, contract terms, and special offers** for these high-discount accounts.

7. Get the complete report of the Gross sales amount for the customer “Atliq Exclusive” for each month . This analysis helps to get an idea of low and high-performing months and take strategic decisions. The final report contains these columns:

Month
Year
Gross sales Amount

“Atliq Exclusive”



Insights:

- For AtliQ Exclusive, November 2020 marked the highest sale and March 2020 marked the lowest gross sales. It’s evident that the lower sales between March to August are because of COVID-19.
- However, it is a very good sign that the sales increased quickly after August and reached the highest level since the last two years in November.

	months	year	gross_sales
▶	September	2019	9.09M
	October	2019	10.38M
	November	2019	15.23M
	December	2019	9.76M
	January	2020	9.58M
	February	2020	8.08M
	March	2020	0.77M
	April	2020	0.80M
	May	2020	1.59M
	June	2020	3.43M
	July	2020	5.15M
	August	2020	5.64M
	September	2020	19.53M
	October	2020	21.02M
	November	2020	32.25M
	December	2020	20.41M
	January	2021	19.57M
	February	2021	15.99M
	March	2021	19.15M
	April	2021	11.48M
	May	2021	19.20M
	June	2021	15.46M
	July	2021	19.04M
	August	2021	11.32M

FY 2020
79.50M

FY 2021
224.42M

8. In which quarter of 2020, got the maximum total_sold_quantity? The final output contains these fields sorted by the total_sold_quantity,

Quarter
total_sold_quantity

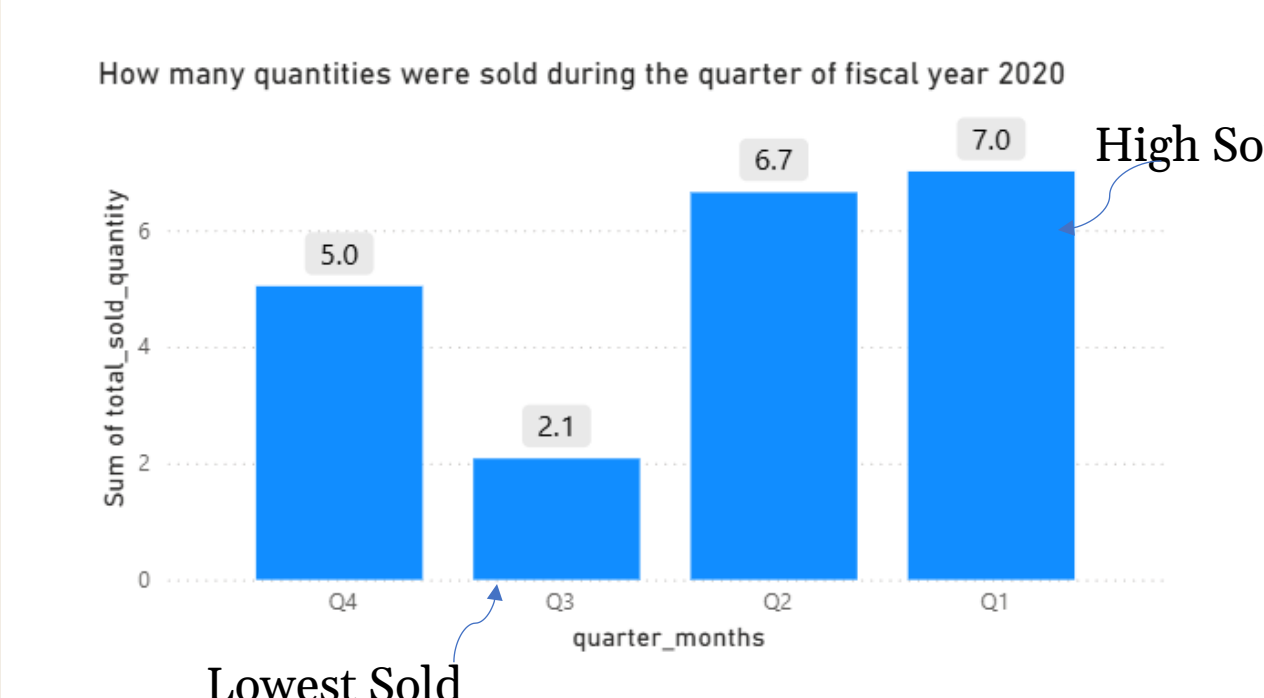
Output

Insights:

	quarter_months	total_sold_quantity
►	Q1	7.01M
	Q2	6.65M
	Q4	5.04M
	Q3	2.08M



- 1. This supports the earlier insight about COVID-19’s impact on sales. The sold quantity dropped to **2.1M in Q3** (March–May), when the pandemic was at its peak.
- 2. But sales picked up quickly in **Q4**, mainly because students and professionals needed more desktops, notebooks, and computer accessories for online classes and work-from-home.



9. Which channel helped to bring more gross sales in the fiscal year 2021 and the percentage of contribution? The final output contains these fields, channel

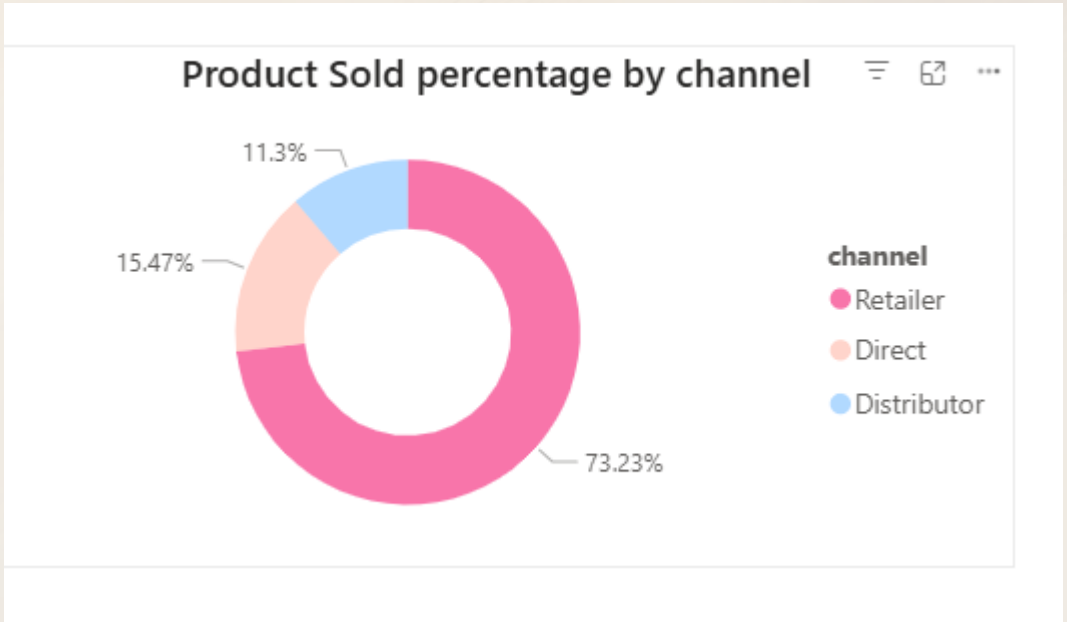
gross_sales_mln
percentage

Insights:

Most of our sales came through **retailers**, making up 75% of the total. Only a **small** share of sales happened through **direct** and **distributor** channels.

Output

	channel	gross_sales_ml	percentage
▶	Retailer	1219.08	73.23%
	Direct	257.53	15.47%
	Distributor	188.03	11.30%



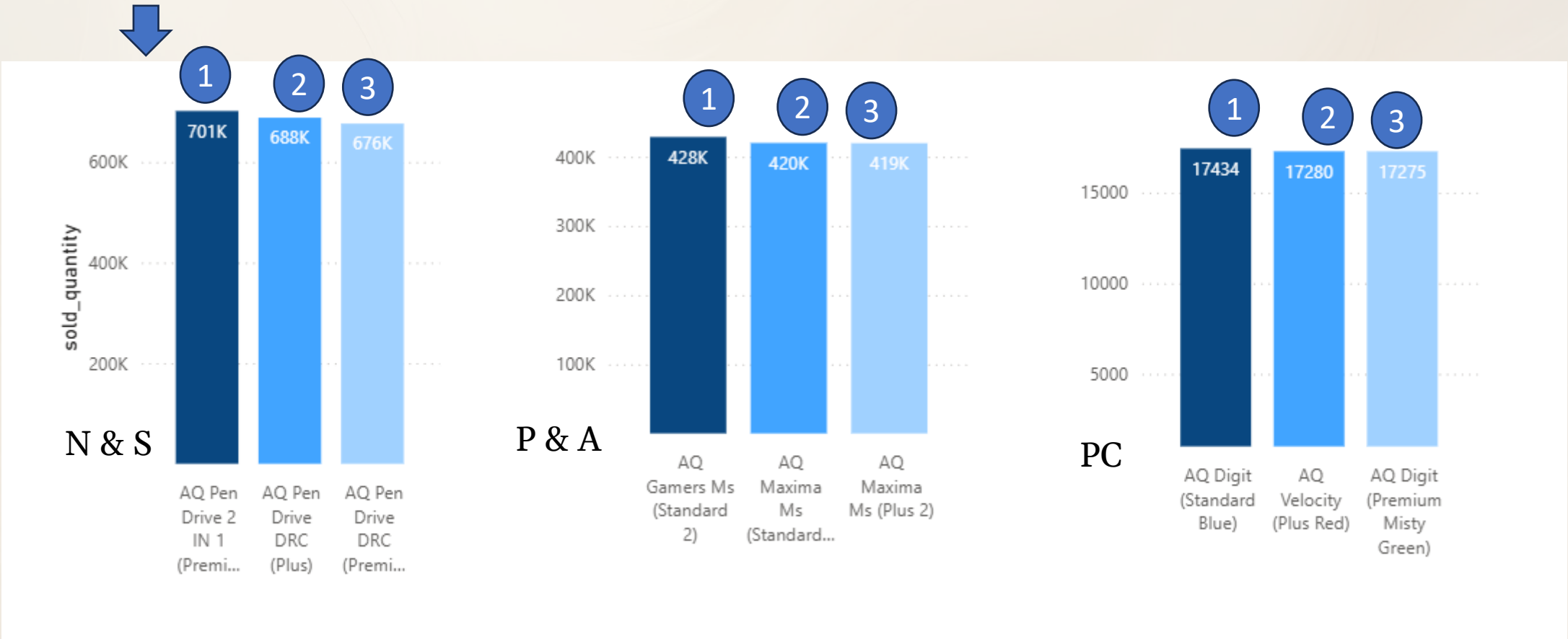
10. Get the Top 3 products in each division that have a high total_sold_quantity in the fiscal_year 2021? The final output contains these fields,

Output

	product_code	product	division	total_sold_quantity	rank_order
▶	A6720160103	AQ Pen Drive 2 IN 1	N & S	701373	1
	A6818160202	AQ Pen Drive DRC	N & S	688003	2
	A6819160203	AQ Pen Drive DRC	N & S	676245	3
	A2319150302	AQ Gamers Ms	P & A	428498	1
	A2520150501	AQ Maxima Ms	P & A	419865	2
	A2520150504	AQ Maxima Ms	P & A	419471	3
	A4218110202	AQ Digit	PC	17434	1
	A4319110306	AQ Velocity	PC	17280	2
	A4218110208	AQ Digit	PC	17275	3

Insights:

- 1. The top-selling product in the N & S category were **Pendrives**, with about **7 lakh units sold**.
- 2. In the Peripherals & Accessories category, the top-selling product was the **mouse**, with around **4 lakh units sold**.
- 3. In the PC category, the highest-selling product was the **personal laptop**, with approximately **1,700 units sold**.



Which are the top 3 selling products in each of Atliq Hardware in FY2021?

END

THANK YOU